NORMATIVE

$T = I$ $\det = +1$ $(x, y) \to (x, y)$

"This is the standard."

RFC 2119 MUST / SHALL

DNS Tool

Communication Standards — Clarity & Vision Quality Gate

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PROJECT dnstool.it-help.tech **SOURCE** github.com/careyjames/dns-tool-web

VERSION 26.37.35 · **LICENSE** BUSL-1.1

Companion to Founder's Manifesto and Philosophical Foundations

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The Owl Semaphore System

The Owl of Athena encodes document classification as semantic state through orientation and color. Each designation is a distance-preserving isometry drawn from the orthogonal group O(2). The system uses three transforms to classify every publication in the DNS Tool corpus.

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NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

$(x, y) \to (-x, y)$

"This reflects the standard."

Informative / Advisory (NOTE)

CRITICAL

$T = C_2 \quad \det = +1$

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Redundant encoding: Color, orientation, and context label provide three independent channels. The semaphore remains distinguishable at small render sizes and in degraded single-channel contexts. Each transform is a distance-preserving isometry in O(2). **Standards mapping:** Normative = RFC 2119 MUST/SHALL. Non-normative = Informative/NOTE. Critical = RFC 2119 MUST NOT/SHALL NOT; CVE/Security advisory.

1. Dual-Gate Requirement

Every piece of user-facing text must pass **both** gates simultaneously. Passing one gate but failing the other is a failure.

Gate 1: Clarity

The words chosen must be clear, communicative, and logical to the reader. Text should educate, not obscure.

REQUIREMENT	LEVEL	MEASUREMENT
Jargon with first use	MUST	Every acronym expanded on first use per page, or linked
Active voice	SHOULD	Prefer "The SPF record authorizes..." over passive
Sentence length	SHOULD	Body copy sentences should not exceed 35 words

One idea per paragraph	SHOULD	Each paragraph should advance one concept
Action-oriented findings	MUST	State what was observed, why it matters, what to do
RFC citation	MUST	Claims about protocol behavior must cite specific RFC section
Transformative language	SHOULD	Prefer language that teaches rather than restates

Gate 2: Vision

The visual presentation must not cause strain or impede comprehension. The reader's eyes matter.

REQUIREMENT	LEVEL	MEASUREMENT
Minimum font size	MUST	No rendered text below 0.75rem (12px)
Body text minimum	SHOULD	Primary reading text at least 0.875rem (14px)
Contrast ratio (normal text)	MUST	WCAG 2.1 AA: minimum 4.5:1 against background
Contrast ratio (large text)	MUST	WCAG 2.1 AA: minimum 3:1 for 18px+ or 14px+ bold
Text overflow	MUST	No text beyond container boundary
Line length	SHOULD	Body text ≤ 80 characters per line
Line height	SHOULD	Body text at least 1.5 line-height
Scotopic mode	MUST	Minimum 4.5:1 contrast against covert background (#0a0c10)
Focus indicators	MUST	Visible focus indicators on all interactive elements

2. Audience Calibration

DNS Tool serves three primary audiences with different reading contexts:

The Engineer

Context: Analyzing a domain at a terminal, possibly at 2 AM, under incident pressure.
Needs: Precision, RFC references, verification commands, machine-parseable output.
Vision concern: Extended reading sessions on dark backgrounds. Font size and contrast directly affect error rate.

The Executive

Context: Reading a brief before a board meeting or vendor assessment.
Needs: Risk posture in business terms, remediation cost/complexity, decision points.
Vision concern: May be reading on phone, tablet, or printed page. Layout must be responsive.

The Older Professional

Context: 20+ years in infosec, may have age-related vision changes (presbyopia, reduced contrast sensitivity).
Needs: Everything both audiences need, plus accommodation for reduced visual acuity.
Vision concern: This is the primary reason the font-size floor exists. 10px text that "looks fine" to a 25-year-old developer may be unreadable to the 55-year-old CISO.

3. Enforcement

Automated Checks

CHECK	TOOL	THRESHOLD
Font size floor	CSS audit	Zero sub-0.75rem values
Contrast ratio	squirrelscan	Zero errors

Text overflow	Manual viewport testing	No horizontal scrollbar at 375px, 768px, 1024px, 1440px
Heading hierarchy	squirrelscan	No skipped levels
Lighthouse accessibility	Chrome Lighthouse	Score ≥ 95

Manual Review Checklist

- **CLARITY** Can a non-technical executive understand the main finding within 10 seconds?
- **CLARITY** Does every acronym have a first-use expansion or link?
- **CLARITY** Does every security finding include a “what to do” action?
- **VISION** At 50cm / 20″, can all text be read on a 13″ laptop at 100% zoom?
- **VISION** In scotopic mode, does all text pass WCAG AA contrast (4.5:1)?
- **VISION** On a 375px mobile viewport, does all content fit without horizontal scroll?
- **VISION** Are interactive elements large enough for touch targets (44×44px)?

4. CSS Reference Values

Enforced values as of v26.37.35:

PROPERTY	VALUE	CONTRAST RATIO	STANDARD
Font size floor	`0.75rem` (12px)	—	All `.u-fs-*` utility classes enforce this minimum
`.text-muted`	`#9ca3af` on `#0d1117`	~7.45:1	WCAG AA (requires 4.5:1)
`.card.bg-dark .text-muted`	`#adb5bd` on `#21262d`	~8.8:1	WCAG AA
Secondary text minimum	`#8b949e` on `#0d1117`	~6.15:1	Replaces prior `#6c757d` (~4.04:1)

5. Connection to Project Philosophy

[ACCOUNTABILITY] — If the text is unreadable, the finding is unaccountable: it exists but cannot be acted upon.

[ZERO_WASTE] — Text that cannot be read is wasted: it consumes space without delivering value.

[VERBOSE+++++] — Clarity does not mean brevity. It means every word earns its place by advancing the reader's understanding.

Vision as Ethics — Causing eye strain to a reader who came for security intelligence is a design failure with real consequences.

OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER

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COMPOSITION IDENTITIES IN O(2)

$\sigma \circ \sigma = I$ · $R(\theta_1) \circ R(\theta_2) = R(\theta_1+\theta_2)$ · $\sigma \circ R(\theta) = \sigma_\theta$

"A score without a citation is an opinion. A finding without readability is invisible."

Related documents: Founder's Manifesto · Philosophical Foundations · Methodology Paper · Reference Library · Sources & Citations